

Marvel Snap Economy Model - Looplab

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Project Description

This project models a simplified version of the Marvel Snap economy using Looplab, the React-based Machinations tool used by the MGI team. The model connects player activity, rewards, card progression, Collection Level, and random match outcomes.

The model connects:

- Player sessions and matches
- Credits and Boosters
- Card upgrades
- Collection Level
- Collection Level rewards
- Tokens and card purchases
- Wins, losses, and Cubes
- Wild Booster rewards
- Result charts and an End Condition

Model Flow

1. A player starts a session.
2. Daily Missions give Credits.
3. Play Matches gives normal Boosters and creates five match attempts.
4. Each match produces either a win or a loss.
5. Wins add a random 1 to 6 Cubes, while losses remove a random 1 to 6 Cubes.
6. Each match also has a chance to give Wild Boosters.
7. Credits and Boosters are spent on card upgrades.
8. Card upgrades increase Collection Level.
9. Collection Level unlocks rewards and new cards.
10. Tokens from Collection Level rewards purchase Common Cards.
11. Charts record the economy, Boosters, progression, and match results.
12. The simulation ends when Sessions Played reaches 150.

Card Upgrade Costs

Upgrade	Credits	Boosters	Collection Level Gain
Common to Uncommon	25	5	+1
Uncommon to Rare	100	10	+2
Rare to Epic	200	20	+4
Epic to Legendary	300	30	+6
Legendary to Ultra	400	40	+8
Ultra to Infinity	500	50	+10

Model Assumptions

- Starting balance: 500 Credits
- Starting collection: 12 Common Cards
- Starting Boosters and Cubes: 0
- Five matches per completed session
- 100 Credits and 18 normal Boosters earned per session
- 55% win chance and 45% loss chance
- Random 1 to 6 Cube gain after a win
- Random 1 to 6 Cube loss after a loss
- 10% Wild Booster chance, with six Boosters per successful event
- One Collection Level reward every 120 levels
- Each Collection Level reward gives 3,000 Tokens, 1,000 Credits, and 100 Boosters
- One new Common Card is unlocked every 50 Collection Levels
- 3,000 Tokens purchase one Common Card
- Boosters are stored in one shared pool
- Upgrades and card purchases happen automatically

Looplab Migration

The original Machinations economy rules were kept, but several event paths were adjusted for Looplab. Match results and Wild Booster rolls now use explicit attempt pools, legacy trigger behavior was replaced with direct converter outputs, and Cube changes use Looplab's supported 1 to 6 connection range. The layout was then cleaned so that each system could be read as a separate section.

Simulation Method

The model was first checked with a Quick Run using seed 1. It was then tested across 100 deterministic runs using seeds 1 through 100. The same Looplab engine was used for the browser model, command-line results, and automated tests.

The following checks were performed:

- The XML imported without warnings.
- The End of Run condition activated at 150 sessions.
- Wins and losses added up to the completed match count.
- Wild Booster results added up to the completed match count.
- Credits, Boosters, Tokens, and Cubes changed correctly.
- Card pools added up to Cards Owned.
- Collection Level rewards and card unlock progress balanced correctly.
- The project tests and production build completed successfully.

Results

Quick Run Results - Seed 1

Metric	Result
End Time	150
Completed Matches	745
Wins	409
Losses	336
Win Rate	54.9%
Wild Bonus Results	79
No Wild Bonus Results	666
Wild Bonus Rate	10.6%
Credits Remaining	75
Boosters Remaining	1,579
Collection Level	359
Cards Owned	21
Cubes Remaining	225
Tokens Remaining	0
CL Reward Progress	119
Card Unlock Progress	9
CL Rewards Claimed	2

Results from 100 Runs

Measurement	Result Across Runs
Runs Reaching End of Run	100 out of 100
Average Run Length	150 steps
Completed Matches	745 per run
Average Win Rate	55.13%
Average Wild Booster Rate	10.20%
Final Credits	75 in every run
Average Final Boosters	1,560.76
Final Booster Range	1,477 to 1,669
Collection Level	359 in every run
Average Final Cubes	279.56
Final Cube Range	35 to 492
Cards Owned	21 in every run
CL Rewards Claimed	2 in every run
CL Reward Progress	119 in every run
Card Unlock Progress	9 in every run
Final Tokens	0 in every run

Validation Results

- All 100 runs reached the End of Run condition.
- Every run ended at exactly 150 sessions.
- Every run resolved 745 matches.
- The average win rate was close to the configured 55%.
- The average Wild Booster rate was close to the configured 10%.
- All three automated model tests passed.
- The Looplab production build passed.
- No XML import warnings were produced.

Why the Main Totals Are Correct

The final card pools contain 1 Rare Card, 8 Legendary Cards, and 12 Ultra Cards. These values add up to 21 Cards Owned. Their Collection Level gains also balance: 12 Ultra Cards produce 252 levels, 8 Legendary Cards produce 104 levels, and 1 Rare Card produces 3 levels. The total is 359 Collection Levels.

At Collection Level 359, two complete 120-level rewards are claimed and 119 progress remains. Seven Common Cards are unlocked through the 50-level rule, leaving 9 Card Unlock Progress. The final card count is 12 starting cards, 7 unlocked cards, and 2 Token purchases, which equals 21 cards.

Analysis of Results

Credits Were the Main Bottleneck

The clearest result was the difference between the final Credit and Booster balances. Every run ended with only 75 Credits, while the average final Booster balance was about 1,561. Every upgrade requires both resources. The player had enough Boosters for more upgrades but could not use them because the Credit balance was too low.

This means that adding more Booster rewards would not significantly increase Collection Level under the current assumptions. Increasing Credit income or reducing Credit costs would have a larger effect.

Collection Progression Was Stable

Collection Level stayed at 359 and Cards Owned stayed at 21 in every run. The stable result occurred because session Credit income, upgrade costs, Collection Level rewards, and card unlock points were fixed. Random match results changed Cubes and Wild Boosters, but they did not change the final progression because Boosters were already above the amount that the available Credits could spend.

Match Results Behaved as Expected

The configured win chance was 55%, and the observed average across 100 runs was 55.13%. The seed 1 Quick Run produced 409 wins and 336 losses, giving a 54.9% win rate. Small differences between individual runs are normal in a random system.

Wild Booster Results Behaved as Expected

The configured Wild Booster chance was 10%, and the 100-run average was 10.20%. The seed 1 Quick Run produced 79 Wild Booster results and 666 No Wild Bonus results, giving a 10.6% Wild Booster rate. This is close to the intended probability.

Cubes Had the Greatest Variation

Final Cubes ranged from 35 to 492. Cubes changed more than progression because they depended on both the random match result and the random 1 to 6 amount gained or lost. This variation shows that repeated simulations were producing different outcomes even though Collection Level remained stable.

Collection Rewards Created a Feedback Loop

Card upgrades increased Collection Level. Collection Level then produced Credits, Boosters, Tokens, and new cards. Those rewards supported further upgrades. The complete loop was: Play Matches, earn resources, upgrade cards, gain Collection Level, claim rewards, receive more resources and cards, and upgrade again.

Zero Tokens at the End Was Expected

Two Collection Level rewards created 6,000 Tokens. The Buy Card converter immediately spent those Tokens on two Common Cards at 3,000 Tokens each. The final Token balance was therefore zero even though Tokens were earned during the simulation.

Why the Model Produced 745 Matches

The simulation stops as soon as Sessions Played reaches 150. The five match attempts created on the final step remain in the Match Attempts pool and are not resolved. This means that 149 sessions are fully completed: 149 completed sessions multiplied by 5 matches equals 745 completed matches. This is expected event timing, not a failure.

Main Finding

The model shows that Credits limit card progression before Boosters. Boosters accumulate faster than the player can spend them, while Credits repeatedly fall close to zero. Collection Level remains stable across repeated runs, while wins, losses, Wild Booster rewards, and Cubes show the expected random variation.

The next useful experiment would increase Credit income by 10% and 20% to measure how much Collection Level increases and whether Boosters eventually become the next progression bottleneck.

What I Learned

Converting a Game Economy Into a System Model

I learned how to separate a game into resources, costs, rewards, progress counters, random outcomes, and player actions. This made it easier to see how one economy change can affect several other systems.

Moving the Model Into Looplab

Moving the model showed me that matching the visible diagram is not enough. Event timing, gate inputs, resource paths, converters, and the End Condition must also behave correctly in the new engine. Rebuilding the legacy trigger paths as explicit Looplab connections made the model easier to test and explain.

Building Interconnected Systems

By connecting Credits, Boosters, Tokens, cards, Collection Level, rewards, and match outcomes, I could observe a complete economy loop instead of separate counters. A small change in Credit income could affect upgrades, rewards, and card ownership downstream.

Identifying a Bottleneck

Looking only at the final Collection Level was not enough. Comparing the remaining Credits and Boosters explained why progression stopped. A large Booster balance with only 75 Credits clearly showed that Credits were limiting additional upgrades.

Why Multiple Runs Are Important

One run shows only one possible random outcome. Running the model 100 times showed which results were stable and which results varied. Collection Level and card ownership were stable. Wins, losses, and Wild Boosters varied slightly, while Cubes showed the greatest variation.

Separating Rules From Assumptions

The model combines upgrade values with simplified assumptions such as a fixed win rate, pooled Boosters, automatic upgrades, automatic card purchases, and simplified Collection Level rewards. Recording these assumptions prevents the model from being presented as an exact copy of the live game.

Testing and Reproducibility

I learned how useful repeatable seeds and automated tests are when validating a random economy. The same seed reproduces the same outcome, while a batch of different seeds checks whether long-term probabilities and resource totals remain reasonable. This was a new and enjoyable experience for me.

Limitations

The model does not include every part of the live Marvel Snap economy.

- Boosters are pooled instead of being card-specific.
- Player card-selection decisions are not modeled.
- Upgrades and card purchases happen automatically.
- Snap and retreat decisions are simplified using random 1 to 6 Cube changes.
- The complete Collection Level reward track is not included.
- Gold, bundles, variants, season passes, and shops are outside the model.
- Matchmaking and deck strength do not change the win rate.
- Live game values may change over time.

Conclusion

The Marvel Snap economy model was successfully moved into Looplab and connected player sessions, match outcomes, resources, upgrades, Collection Level, rewards, Tokens, card acquisition, Wild Boosters, Cubes, and result charts.

All 100 repeated simulations reached the End of Run condition at 150 sessions. The average 55.13% win rate was close to the configured 55%, and the average 10.20% Wild Booster rate was close to the configured 10%. The final Collection Level, card count, reward progress, and Token balance were internally consistent in every run.

The most important result was that Credits became the main progression bottleneck while Boosters accumulated. Collection Level remained stable, while match outcomes and Cubes showed expected random variation.

The model now provides a repeatable foundation for testing future economy changes such as Credit income, upgrade costs, match volume, reward frequency, and player spending behavior in Looplab.